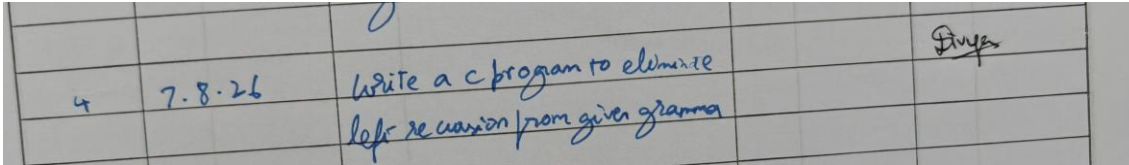


Assessment – 4

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a)

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main() {
    int n;
    char production[100];
    printf("Enter the number of productions: ");
    scanf("%d", &n);
    getchar();
    while (n--) {
        char alpha[10][50], beta[10][50];
        int a = 0, b = 0;
        printf("Enter production: ");
        fgets(production, sizeof(production), stdin);
        production[strcspn(production, "\n")] = '\0';
        char temp[100];
        int j = 0;
        for (int i = 0; production[i] != '\0'; i++) {
            if (!isspace((unsigned char)production[i]))
                temp[j++] = production[i];
        }
        temp[j] = '\0';
        char lhs = temp[0];
        char *rhs = strstr(temp, "->");
        if (rhs == NULL) {
            printf("Invalid production!\n");
        }
    }
}
```

```

continue;
}

rhs += 2;

char rhsCopy[100];

strcpy(rhsCopy, rhs);

char *token = strtok(rhsCopy, "|");

while (token != NULL) {

if (token[0] == lhs) {

strcpy(alpha[a++], token + 1);

} else {

strcpy(beta[b++], token);

}

token = strtok(NULL, "|");

}

if (a == 0) {

printf("%c -> %s\n", lhs, rhs);

} else {

printf("%c -> ", lhs);

for (int i = 0; i < b; i++) {

printf("%s%c", beta[i], lhs);

if (i != b - 1)

printf(" | ");

}

printf("\n");

printf("%c' -> ", lhs);

for (int i = 0; i < a; i++) {

printf("%s%c", alpha[i], lhs);

if (i != a - 1)

printf(" | ");

}

printf(" | ε\n");

}

printf("24BCE2739\n");

}

```

```
return 0;
```

```
}
```

```
anaconda@sjt418scope061:~/24bce2739$ ./leftrecurse
Enter the number of productions: 3
Enter production: E->T'E'
E -> T'E'
24BCE2739
Enter production: T->T*F|F
T -> FT'
T' -> *FT' | ε
24BCE2739
Enter production: F->id|(E)
F -> id|(E)
24BCE2739
```

```
anaconda@sjt418scope061:~/24bce2739$ ./leftrecurse
Enter the number of productions: 2
Enter production: S ->(L)|a
S -> (L)|a
24BCE2739
Enter production: L -> L,S | S
L -> SL'
L' -> ,SL' | ε
24BCE2739
```

B)

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main() {
```

```
int n, i, j, k;
```

```
char lhs;
```

```
char prod[20][50];
```

```
printf("Enter non terminal: ");
```

```
scanf(" %c", &lhs);
```

```
printf("Enter number of productions: ");
```

```
scanf("%d", &n);
```

```
printf("Enter productions:\n");
```

```
for (int i=0;i<n;i++) {
```

```
scanf("%s", prod[i]);
```

```
}
```

```
char prefix[50];
```

```

strcpy(prefix, prod[0]);

for (i =1;i<n;i++) {

k = 0;

while (prefix[k] != '\0' && prod[i][k] != '\0' && prefix[k] == prod[i][k]) {

k++;

}

prefix[k] = '\0';

}

if (strlen(prefix) == 0) {

printf("\nno left factoring required\n");

printf("%c -> ", lhs);

for (int i =0;i<n;i++) {

printf("%s", prod[i]);

if (i != n-1){

printf(" | ");

}

}

printf("\n");

return 0;

}

printf("\nAfter left factoring\n");

printf("%c -> %s%c\n", lhs, prefix, lhs);

printf("%c' -> ", lhs);

for (int i =0;i<n;i++) {

char *remaining = prod[i] + strlen(prefix);

if (strlen(remaining) == 0) {

printf("e");

}

else {

printf("%s", remaining);

}

if (i != n-1){

printf(" | ");

}

}

```

```
}  
printf("\n");  
return 0;  
}
```

```
anaconda@sjt418scope066:~/24bce2739$ ./leftfact  
Enter non terminal: A  
Enter number of productions: 3  
Enter productions:  
abcd  
abef  
abxy  
  
After left factoring  
A -> abA'  
A' -> cd | ef | xy
```

```
Enter non terminal: B  
Enter number of productions: 3  
Enter productions:  
cAB  
cb  
cab  
  
After left factoring  
B -> cB'  
B' -> AB | b | ab
```